



National
Qualifications
2022

X857/76/11

Physics
Paper 2 — Relationships sheet

FRIDAY, 13 MAY

10:15 AM – 12:30 PM



* X 8 5 7 7 6 1 1 *

Relationships required for Physics Higher

$$d = \bar{v}t$$

$$s = \bar{v}t$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$s = \frac{1}{2}(u + v)t$$

$$F = ma$$

$$W = mg$$

$$E_w = Fd, \text{ or } W = Fd$$

$$E_p = mgh$$

$$E_k = \frac{1}{2}mv^2$$

$$P = \frac{E}{t}$$

$$p = mv$$

$$Ft = mv - mu$$

$$F = G \frac{m_1 m_2}{r^2}$$

$$t' = \frac{t}{\sqrt{1 - \left(\frac{v}{c}\right)^2}}$$

$$l' = l \sqrt{1 - \left(\frac{v}{c}\right)^2}$$

$$f_o = f_s \left(\frac{v}{v \pm v_s} \right)$$

$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}}$$

$$z = \frac{v}{c}$$

$$v = H_0 d$$

$$W = QV$$

$$E = mc^2$$

$$I = \frac{P}{A}$$

$$I = \frac{k}{d^2}$$

$$I_1 d_1^2 = I_2 d_2^2$$

$$E = hf$$

$$E_k = hf - hf_0$$

$$v = f\lambda$$

$$E_2 - E_1 = hf$$

$$d \sin \theta = m\lambda$$

$$n = \frac{\sin \theta_1}{\sin \theta_2}$$

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2}$$

$$\sin \theta_c = \frac{1}{n}$$

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$$

$$I_{rms} = \frac{I_{peak}}{\sqrt{2}}$$

$$T = \frac{1}{f}$$

$$V = IR$$

$$P = IV = I^2 R = \frac{V^2}{R}$$

$$R_T = R_1 + R_2 + \dots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$V_1 = \left(\frac{R_1}{R_1 + R_2} \right) V_S$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$E = V + Ir$$

$$C = \frac{Q}{V}$$

$$Q = It$$

$$E = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2} \frac{Q^2}{C}$$

$$\text{path difference} = m\lambda \quad \text{or} \quad \left(m + \frac{1}{2}\right)\lambda \quad \text{where } m = 0, 1, 2, \dots$$

$$\text{random uncertainty} = \frac{\text{max. value} - \text{min. value}}{\text{number of values}}$$

or

$$\Delta R = \frac{R_{\text{max}} - R_{\text{min}}}{n}$$

Additional relationships

Circle

$$\text{circumference} = 2\pi r$$

$$\text{area} = \pi r^2$$

Sphere

$$\text{area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Electron arrangements of elements

	Group 1	Group 2
1	1	1
2	1	1
3	1	1
4	1	1
5	1	1
6	1	1
7	1	1
8	1	1
9	1	1
10	1	1
11	1	1
12	1	1
13	1	1
14	1	1
15	1	1
16	1	1
17	1	1
18	1	1
19	1	1
20	1	1
21	1	1
22	1	1
23	1	1
24	1	1
25	1	1
26	1	1
27	1	1
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79	1	1
80	1	1
81	1	1
82	1	1
83	1	1
84	1	1
85	1	1
86	1	1
87	1	1
88	1	1
89	1	1
90	1	1
91	1	1
92	1	1
93	1	1
94	1	1
95	1	1
96	1	1
97	1	1
98	1	1
99	1	1
100	1	1

(1)

1 H 1	(2)
Hydrogen	
3 Li 2,1	4 Be 2,2
Lithium	Beryllium

Key

Atomic number	Symbol	Electron arrangement	Name
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Transition elements

Sodium	Magnesium	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
2,8,8,1	2,8,8,2	2,8,9,2	2,8,10,2	2,8,11,2	2,8,13,1	2,8,13,2	2,8,14,2	2,8,15,2	2,8,16,2	2,8,18,1	2,8,18,2
Potassium	Calcium	Scandium	Titanium	Vanadium	Chromium	Manganese	Iron	Cobalt	Nickel	Copper	Zinc
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd
2,8,18,8,1	2,8,18,8,2	2,8,18,9,2	2,8,18,10,2	2,8,18,12,1	2,8,18,13,1	2,8,18,13,2	2,8,18,15,1	2,8,18,16,1	2,8,18,18,0	2,8,18,18,1	2,8,18,18,2
Rubidium	Strontium	Yttrium	Zirconium	Niobium	Molybdenum	Technetium	Ruthenium	Rhodium	Palladium	Silver	Cadmium
55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg
2,8,18,18,8,1	2,8,18,18,8,2	2,8,18,18,9,2	2,8,18,32,10,2	2,8,18,32,11,2	2,8,18,32,12,2	2,8,18,32,13,2	2,8,18,32,14,2	2,8,18,32,15,2	2,8,18,32,17,1	2,8,18,32,18,1	2,8,18,32,18,2
Cesium	Barium	Lanthanum	Hafnium	Tantalum	Tungsten	Rhenium	Osmium	Iridium	Platinum	Gold	Mercury
87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn
2,8,18,32,18,8,1	2,8,18,32,18,8,2	2,8,18,32,18,9,2	2,8,18,32,32,10,2	2,8,18,32,32,11,2	2,8,18,32,32,12,2	2,8,18,32,32,13,2	2,8,18,32,32,14,2	2,8,18,32,32,15,2	2,8,18,32,32,17,1	2,8,18,32,32,18,1	2,8,18,32,32,18,2
Francium	Radium	Actinium	Rutherfordium	Dubnium	Seaborgium	Bohrium	Hassium	Meitnerium	Darmstadtium	Roentgenium	Copernicium

Lanthanides

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
2,8,18, 18,9,2	2,8,18, 20,8,2	2,8,18,21, 8,2	2,8,18,22, 8,2	2,8,18,23, 8,2	2,8,18,24, 8,2	2,8,18,25, 8,2	2,8,18,25, 9,2	2,8,18,27, 8,2	2,8,18,28, 8,2	2,8,18,29, 8,2	2,8,18,30, 8,2	2,8,18,31, 8,2	2,8,18,32, 8,2	2,8,18,32, 9,2
Lanthanum	Cerium	Praseodymium	Neodymium	Promethium	Samarium	Europium	Gadolinium	Terbium	Dysprosium	Holmium	Erbium	Thulium	Ytterbium	Lutetium

Actinides

89	Ac	2,8,18,32,18,9,2
90	Th	2,8,18,32,18,10,2
91	Pa	2,8,18,32,20,9,2
92	U	2,8,18,32,21,9,2
93	Np	2,8,18,32,22,9,2
94	Pu	2,8,18,32,24,8,2
95	Am	2,8,18,32,25,8,2
96	Cm	2,8,18,32,25,9,2
97	Bk	2,8,18,32,27,8,2
98	Cf	2,8,18,32,28,8,2
99	Es	2,8,18,32,29,8,2
100	Fm	2,8,18,32,30,8,2
101	Md	2,8,18,32,31,8,2
102	No	2,8,18,32,32,8,2
103	Lr	2,8,18,32,32,9,2
Actinium		
Thorium		
Protactinium		
Uranium		
Neptunium		
Plutonium		
Americium		
Curium		
Berkelium		
Californium		
Einsteinium		
Fermium		
Mendelevium		
Nobelium		
Lawrencium		

2 He 2					
Helium	(13)	(14)	(15)	(16)	(17)
5 B 2,3	6 C 2,4	7 N 2,5	8 O 2,6	9 F 2,7	10 Ne 2,8
Boron	Carbon	Nitrogen	Oxygen	Fluorine	Neon
13 Al 2,8,3	14 Si 2,8,4	15 P 2,8,5	16 S 2,8,6	17 Cl 2,8,7	18 Ar 2,8,8
Aluminium	Silicon	Phosphorus	Sulfur	Chlorine	Argon
31 Ga 2,8,18,3	32 Ge 2,8,18,4	33 As 2,8,18,5	34 Se 2,8,18,6	35 Br 2,8,18,7	36 Kr 2,8,18,8
Gallium	Germanium	Arsenic	Selenium	Bromine	Krypton
49 In 2,8,18,18,3	50 Sn 2,8,18,18,4	51 Sb 2,8,18,18,5	52 Te 2,8,18,18,6	53 I 2,8,18,18,7	54 Xe 2,8,18,18,8
Indium	Tin	Antimony	Tellurium	Iodine	Xenon
81 Tl 2,8,18,32,18,3	82 Pb 2,8,18,32,18,4	83 Bi 2,8,18,32,18,5	84 Po 2,8,18,32,18,6	85 At 2,8,18,32,18,7	86 Rn 2,8,18,32,18,8
Thallium	Lead	Bismuth	Polonium	Astatine	Radon

(18)

Group 3 Group 4 Group 5 Group 6 Group 7 Group 0